

Blockchain Honor Minor Degree : Sem VIII

HBCC801 : Decentralized Finance (DeFi)

Module - 5 : DeFi Deep Dive

(9 Hours)

Instructors : Lifna C S

Topics to be covered

1. **Maker DAO:** Maker Protocol: Dai Stablecoins, Maker Vaults, Maker Protocol Auctions Maker Actors: Keepers, Price Oracles, Emergency Oracles, DAO Teams, Dai Savings Rate Dai Use case Benefits and Examples
2. **UniSwap:** UniSwap Protocol Overview: How UniSwap Works, EcoSystem Participants, Smart Contracts UniSwap Core Concepts: Swaps, Pools, Flash Swaps, Oracles
3. **Compound:** Compound Protocol: Supplying Assets, Borrowing Assets, Interest Rate Model Compound Implementation and Architecture: cToken Contracts, Interest Rate Mechanics, Borrowing, Liquidation, Price Feeds, Comptroller, Governance
4. **wBTC:** Need for wBTC, Tokenization and common Issues wBTC, Implementation and Technology: Users, Custodian Wallet Setup, Minting, Burning wBTC Governance, wBTC vs Atomic Swaps, Fees, Legal Binding, Trust Model and Transparency

1. Maker DAO

- Maker Protocol:
- Dai Stablecoins,
- Maker Vaults,
- Maker Protocol Auctions
- Maker Actors:
 - Keepers,
 - Price Oracles,
 - Emergency Oracles,
 - DAO Teams,
- Dai Savings Rate (DSR)
- Dai Use case Benefits and Examples



1. Maker DAO

- A decentralized autonomous organization (DAO) built on Ethereum
- Enables **decentralized lending**
- No reliance on centralized institutions
- **DAI** is a crypto-backed stablecoin
- Risk managed through:
 - Collateral ratios
 - Liquidation mechanisms
- Foundation of DeFi
- Offers :
 - Trustless finance
 - Smart contract automation
- Critical building block for future financial systems

Real-World Analogy

- Similar to **Mortgage loan** via property
- Key difference:
 - Fully decentralized
 - No banks involved

1. Maker DAO

Two-Token Model



MKR Token

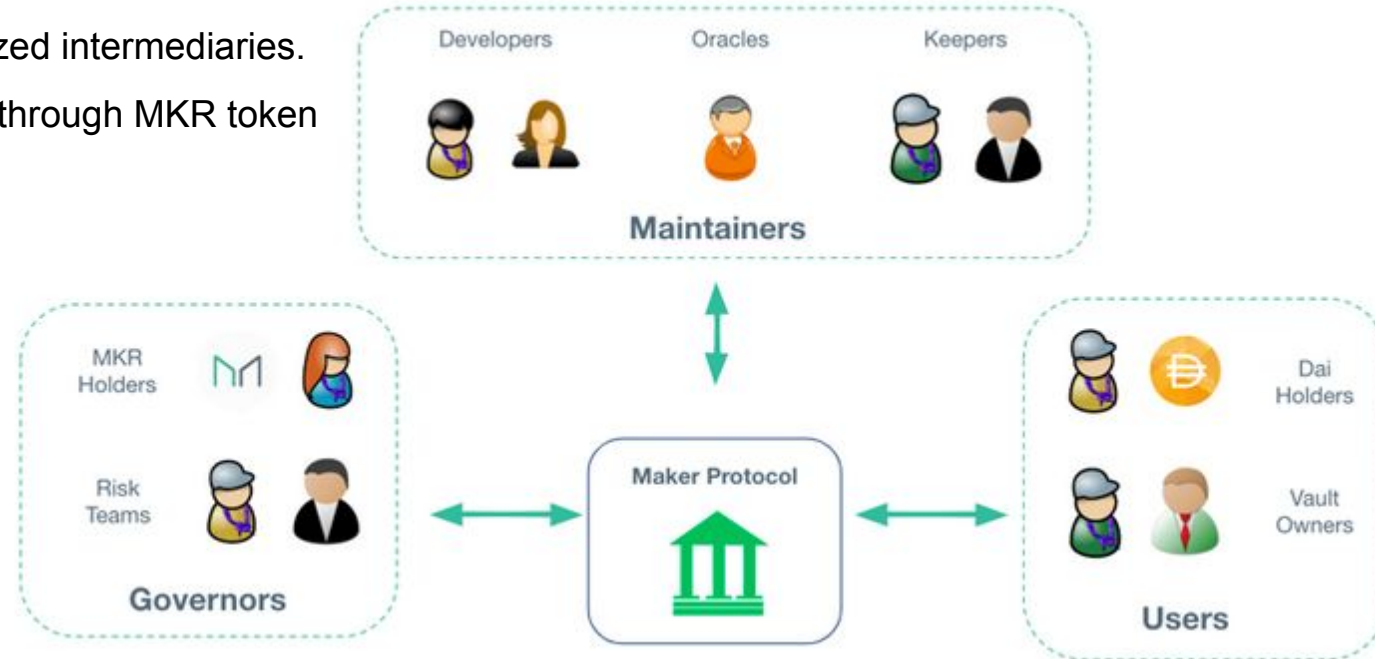
- Governance token
- Voting rights
- Risk management

DAI Token

- pegged to USD
- Widely used across DeFi protocols

1. Maker DAO - Maker Protocol

- Foundational projects in DeFi
- Built on the Ethereum blockchain.
- Enables users to generate the stablecoin DAI.
- Operates without centralized intermediaries.
- Governed by MakerDAO through MKR token holders.



Courtesy : [Maker DAO](https://makerdao.com)

1. Maker DAO - Maker Protocol : System Interaction Actors

User

- Dai Holders

- Basic - No additional knowledge to own Dai
- Stability Seekers; Consumer; Businesses

- Vault Owners

- Intermediate - Some knowledge
- Risk Seekers; Speculators; Borrowers

Governor

- MKR Holders

- Advanced - Extensive knowledge
- Monitor, partake, and vote on upgrades/changes in the Maker Protocol

- Risk Teams

- Advanced - Extensive knowledge
- Collect/compile relevant data and develop risk models, assessed by MKR Holders

1. Maker DAO - Maker Protocol : System Interaction Actors

Maintainer

- **Developers**
 - Advanced - Extensive knowledge
 - Supports system upgrades accepted by Governors
- **Oracles**
 - Intermediate User - Some knowledge
 - Supports data feed from real world to blockchain
- **Keepers**
 - Advanced User - Extensive knowledge
 - Builds/maintains systems that profit off system discrepancies and participate in auctions

Anyone with the required knowledge can freely participate in any role

Any one person (or service) can have multiple roles



1. Maker DAO - Maker Protocol : System Line of Defense

At any point, the system must have more value in collateral than value of Dai supply.

The following mechanisms help maintain system solvency:

1. Supply and Demand

Supply and demand of Vaults (and thus Dai) is influenced by the Stability Fees, Dai Savings Rate, and Debt Ceiling adjustments.

2. Liquidation

Any time the collateral value of a Vault gets closer to its debt, it becomes "risky-er". The system liquidates Vaults that get too risky.

1. Maker DAO - Maker Protocol : System Line of Defense

3. MKR Minting/Burning

If MKR holders govern the Maker Protocol successfully, the Protocol will accrue Surplus Dai as Dai holders pay Stability Fees. On the other hand, if liquidations are inadequate, then the Protocol will accrue Bad Debt. Once this Surplus Dai / Bad Debt amount hits a threshold, as voted by MKR holders, then the Protocol will discharge Surplus Dai / Bad Debt through the Flapper / Flopper smart contract by buying and burning / minting and selling MKR, respectively.

4. Emergency Shutdown

This is a process that is used as a last resort in cases of extreme market irrationality, attacks, or coordinated upgrades. Emergency Shutdown gracefully settles the Maker Platform while ensuring that all users, both Dai holders and Vault users, receive the net value of assets they are entitled to.

1. Maker DAO - Maker Protocol : SAI Vs DAI

Single Collateral DAI

- Single collateral type backing Dai
- Collateral is ETH
- Less diversified
- Liquidations occur at a fixed discount to the current collateral price
- MKR used to pay stability fee
- Decentralized governance through MKR voting rights

Multi Collateral DAI

- Multi. collateral types backing Dai
- Collateral can be any approved asset
- More diversified and stable
- Liquidations are handled using Auctions
- Dai used to pay stability fee
- Decentralized governance through MKR voting rights

Mapping of Terminology ⇒

Single Collateral DAI		Multi Collateral DAI
• Dai Credit System	→	• Maker Protocol
• CDP	→	• (Maker) Vault
• Boom / Bust Spread	→	• Discount / Premium Spread

1. Maker DAO

- **What is a Vault?**
 - Smart contract that:
 - Locks collateral (ETH / ERC-20 tokens)
 - Tracks USD value
 - Users deposit assets → generate DAI
 - Represents a Collateralized Debt Position (CDP)
- **Key Concept – Collateralization**
 - Users must overcollateralize loans
 - Typical ratio: 150% – 200%
 - Ensures system stability despite volatility

1. Maker DAO : Maker Protocol Auctions

- Critical component of the MakerDAO Maker Protocol ecosystem
- Auctions maintain
 - Stability of DAI
 - Protocol solvency
 - Risk management
 - System recapitalization
- Maker Protocol uses **automated smart-contract-based auctions**
 1. Liquidate risky vaults
 2. Manage excess DAI
 3. Cover protocol debt

Auction Type	Purpose
Collateral Auction	Liquidate undercollateralized vaults
Surplus Auction	Sell excess DAI for MKR
Debt Auction	Mint and sell MKR to cover bad debt

1. Maker DAO : Maker Protocol Auctions

1. Collateral Auctions

- Triggered when a vault becomes undercollateralized.
- Process
 1. User deposits collateral (ETH/ERC-20)
 2. Generates DAI
 3. Collateral value drops
 4. Vault falls below liquidation ratio
 5. Keeper initiates auction
 6. Collateral sold for DAI
 7. Debt repaid
- Ex : **User deposits 5 ETH & Borrows 500 DAI**
 - ETH price crashes
 - Vault becomes unsafe
 - Auction Outcomes
 - Portion of ETH sold
 - DAI collected to repay debt
 - Remaining ETH returned to user

1. Maker DAO : Maker Protocol Auctions

2. Surplus Auctions

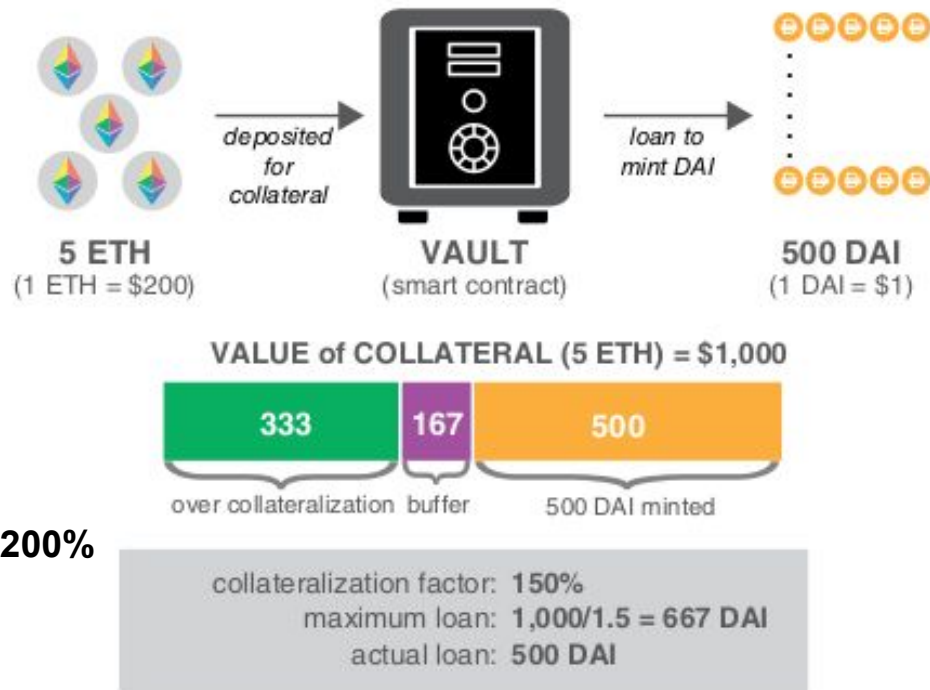
- Occurs when Maker Protocol accumulates excess DAI from: Stability fees / Liquidation penalties
- Objective : Reduce MKR supply.
- Process
 1. Protocol collects surplus DAI
 2. DAI auction initiated
 3. Participants bid MKR
 4. Winning MKR is burned
- Result
 - MKR supply decreases
 - Potential increase in MKR value

3. Debt Auctions

- Triggered when:
 - Collateral liquidation cannot fully cover debt
 - Protocol accumulates bad debt
- Process
 1. Protocol mints new MKR tokens
 2. MKR auctioned for DAI
 3. DAI used to cover debt
- Consequence
 - MKR dilution
 - Governance holders bear risk

1. Maker DAO - Example Scenario

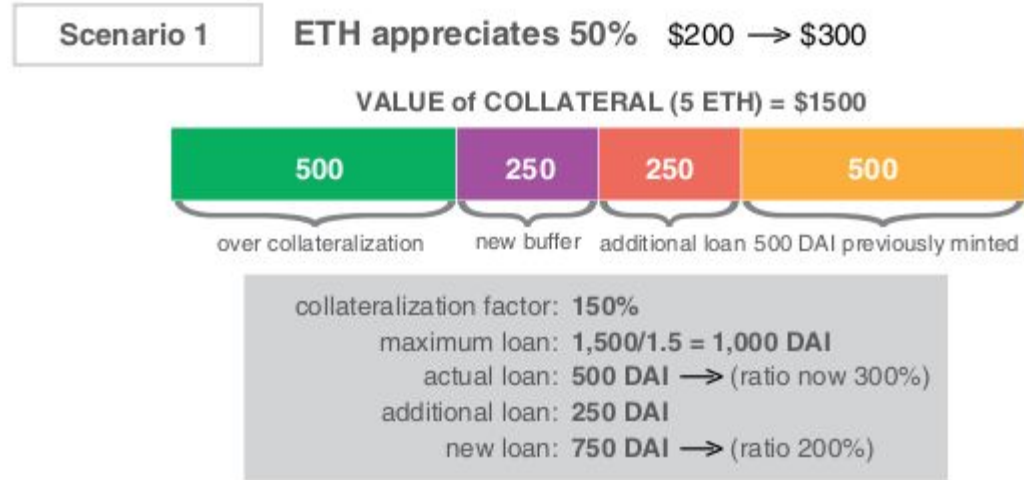
- Investor holds:
 - 5 ETH @ \$200 each = \$1000
- Collateralization ratio: 150%
- Maximum DAI mintable:
 - $DAI = 1000 / 1.5 = 667$
- Risk Buffer Strategy
 - Minting full 667 DAI is risky
- Safer approach:
 - Mint 500 DAI
 - Collateral ratio becomes: $1000 / 500 = 200\%$
- Provides safety margin against price drops



1. Maker DAO - Example Scenario

Scenario 1 – Price Increase

- ETH price rises by 50%
 - New value = \$1500
- Maintain 200% ratio:
 - New DAI = $1500 / 2 = 750 \text{ DAI}$
- Additional DAI mintable = 250 DAI
- Opportunity to increase borrowing



1. Maker DAO - Example Scenario

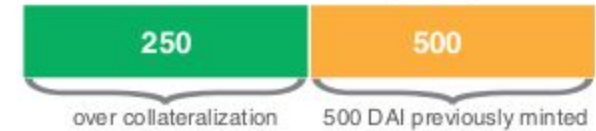
Scenario 2 – Price Decrease

- ETH price drops by 25%
 - New value = \$750
- Collateral ratio becomes: $750 / 500 = 150\%$
- **Reaches liquidation threshold**
- Options for Vault Holder
 1. Add Collateral : Deposit more ETH
 2. Repay Loan : Return 500 DAI
 3. Do Nothing → Liquidation
 - Risk losing collateral

Scenario 2

ETH depreciates 25% \$200 → \$150

VALUE of COLLATERAL (5 ETH) = \$750



collateralization factor: **150%**
 maximum loan: $750 / 1.5 = 500$ DAI
 actual loan: **500 DAI** → (ratio now 150%)

1. Maker DAO - Example Scenario

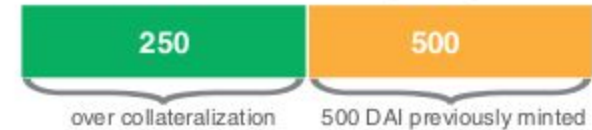
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1. Maker DAO - Example Scenario

Liquidation Process

- Triggered automatically (no warning)
- Executed by keepers (external actors)
- ETH is auctioned to recover DAI
- Ex :
 - 3.33 ETH sold
 - 1.47 ETH returned
 - Keeper earns incentive (0.2 ETH)

Outcome After Liquidation

- User receives:
 - 500 DAI (\$500)
 - 1.47 ETH (~\$220)
- Loss due to: Price drop & Liquidation penalty

Stability Mechanisms

- Overcollateralization
- Market-driven liquidations
- Demand for DAI increases during liquidation
- Helps maintain DAI $\approx$ \$1 peg

1. Maker DAO Comparison

Parameter	Traditional Finance	MakerDAO
Control & Governance	Interest rates and financial policies are controlled by central banks, governments, and financial institutions.	Governed openly by MKR token holders through decentralized voting.
Access to Loans	Loan approval depends on credit history, regulations, documentation, and institutional policies.	Anyone can generate DAI by depositing supported crypto collateral in a vault.
Financial Inclusion	Many individuals lack access to banking and lending services.	Open and permissionless participation for anyone with internet and crypto assets.
Loan Processing Speed	Loan approval may take days or weeks due to verification procedures.	Instant liquidity through smart contracts with minimal delay.
Transaction Costs	High operational and intermediary costs.	Lower transaction costs through blockchain automation.

1. Maker DAO Comparison

Parameter	Traditional Finance	MakerDAO
Interoperability	Traditional USD cannot be directly integrated into smart contracts.	DAI can be seamlessly used across DeFi protocols and smart contracts.
Transparency	Limited visibility into institutional reserves and collateralization.	Vault collateralization ratios are publicly visible on the blockchain.
Collateral Management	Managed by centralized financial institutions.	Managed automatically through decentralized smart contracts.
Liquidity Access	Requires intermediaries such as banks and lenders.	Users can access liquidity directly from the protocol.
Automation	Relies heavily on manual verification and administrative processes.	Fully automated using blockchain-based smart contracts.
Global Accessibility	Access varies by country, regulations, and banking infrastructure.	Globally accessible without geographic restrictions.

1. Maker DAO Comparison

Parameter	Traditional Finance	MakerDAO
Trust Model	Requires trust in banks and centralized institutions.	Trust minimized through transparent blockchain protocols.
Stable Currency Mechanism	Fiat currency stability maintained by central banks and monetary policy.	DAI stability maintained through overcollateralization and governance mechanisms.
Availability	Banking systems operate within business hours and jurisdictions.	Available 24/7 on decentralized networks.
Asset Utilization	Assets often remain idle unless processed through financial institutions.	Crypto assets can generate liquidity without being sold.

1. Maker DAO

Advantages of Maker Protocol

- Decentralized stablecoin
- No dependence on banks
- Transparency
- Automated smart contracts
- Global accessibility
- Liquidity without selling assets

Limitations of Maker Protocol

- Crypto market volatility
- High collateral requirements
- Liquidation risk
- Gas fees on Ethereum
- Peg maintenance challenges

1. Maker DAO Rates

- **Risk Premium Rate** - This rate is used to calculate the risk premium fee that accrues on debt in a Vault. A unique Risk Premium Rate is assigned to each collateral type. (e.g. 2.5%/year for Collateral A, 3.5%/year for Collateral B, etc)
- **Base Rate** - This rate is used to calculate the base fee that accrues on debt in a Vault. A system wide Base Rate is assigned to all collateral types. (e.g. 0.5%/year for the Maker Protocol)
- **Stability Rate** = *Risk Premium Rate + Base Rate*. This rate is used to calculate the **Stability Fee**.
- **Dai Savings Rate (DSR)** - This rate is used to calculate the dai earned that accrues on Dai locked in the savings contract. A system wide Dai Savings Rate is assigned to all Dai locked in the DSR contract. (e.g. 1%/year for DSR)

All rates accrue regularly on a per second basis.